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MILITARY SPECIFICATION

KEYSEATING MACHINES, MECHANICAL AND HYDRAULIC, RECIPROCATING CUTTER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical and hydraulic keyseating machines. The type machines covered herein have a reciprocating cutter, and are used to machine keyseats for square and flat parallel side keys and tapered keys in the bores of gears, propellers, and similar parts.

1.2 Classification. This specification covers keyseating machines of the following types, classes, and sizes. The sizes shown in the classification table below represent the maximum width keyseat that can be machined by each machine. The type, class, and size machine required for a specific procurement shall be as specified (see 6.2.1).

Type I - Mechanical

Class 1 - Manual Feed
Size 3/4 - 3/4 inch (19.05 mm) maximum keyseat width
Size 1 1/2 - 1 1/2 inches (38.1 mm) maximum keyseat width

Class 2 - Power Feed
Size 1 1/4 -1 1/4 inches (31.75 mm) maximum keyseat width
Size 1 1/2 -1 1/2 inches (38.1 mm) maximum keyseat width

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3419

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Type II - Hydraulic

Size 1 1/4 - 1 1/4 inches (31.75 mm) maximum keyseat width
Size 2 - 2 inches (50.8 mm) maximum keyseat width
Size 3 - 3 inches (76.2 mm) maximum keyseat width
Size 3 1/2 - 3 1/2 inches (88.9 mm) maximum keyseat width

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commerical Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B17.1 - Keys and Keyseats
C113.1 - Electrical Standards for Metalworking Machine Tools (NFPA No. 79)
Z210.1 - Metric Practice (ASTM-E-380)

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018).

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 6.2.1 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of machining keyseats in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary Systems of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical

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composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics, and elastomers that have been collected from discarded solid; liquid, semi-solid, or gaseous waste such as garbage, refuse, or sludge.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commerical practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating position Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature wear.

3.4 Components. The machine shall consist essentially of a base, worktable, workpiece centering and hold-down devices, cutter bars, cutter bar drive mechanism, a cutter bar or table feed mechanism, and necessary supports to prevent deflection of the cutter bar while the machine is in operation.

3.4.1 Base. The base shall be either a cast or fabricated structure containing the cutter bar drive mechanism and supporting the worktable and all components mounted thereon. The base shall have sufficient strength and rigidity to support the working parts of the machine and a workpiece not exceeding the maximum workpiece weight specified in Table I, and maintain all parts in alignment to the degree necessary to meet the performance requirements specified herein. The base shall have ample covered access openings to permit access to the interior for cleaning, servicing, and for making necessary adjustments and tool changes. The base shall have means

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for bolting the machine to a foundation. If the machine requires an excavated pit, the pit depth shall not exceed the requirements of Table I.

3.4.2 Worktable. The working surfaces of the worktable shall be machined and shall have provisions for mounting such equipment as workpiece centering devices, center posts, sliding angle blocks, V-blocks, and auxiliary tilting or index tables when the machine design requires such equipment for operation. If the feed mechanism feeds the table instead of the cutter bar, the table shall be mounted on precision ways which may be of the integrally machined type or of the replaceable round or flat bar type. If the table slides on a way, means shall be provided for compensating for wear in the way or way slide. When capabilities of cutting tapered keyseats are required (see 3.4.8), the worktable itself may tilt or an auxiliary tilt type worktable that can readily be attached to or removed from the worktable shall be furnished. In either event, the tilting worktable shall have a dial or scale graduated in "inch per foot" to indicate the tilt angle and shall have means for secure clamping in any position within its tilt range.

3.4.3 Workpiece centering and hold-down devices. The machine shall have devices that will self-center the workpiece bore in proper relationship to the cutter and securely hold the workpiece in proper position during the keyseating operation. The centering method shall be sufficiently accurate to enable the machine to machine keyseats conforming to the accuracy requirements of 3.6.2. Workpiece centering may be accomplished by centering bushings that mount over a center post or by centering bushings or centering jaws that mount onto the worktable. The centering devices shall be capable of accurately centering workpieces having an outside diameter that is concentric with the bore of the workpiece. When specified (see 6.2.1), the centering device shall also be capable of centering workpieces having an outside diameter that is non-concentric with the bore of the workpiece.

3.4.4 Cutter bar and cutter. The cutter bar shall be either of the single or multiple tooth type either having teeth that are cut directly into the cutter bar or having replaceable tooth inserts. Cutter bars and tooth inserts, as applicable, shall be available from the manufacturer for cutting the range of keyseat widths specified in Table I. The cutter bar shall be mounted vertically and shall protrude and reciprocate through an opening in the worktable, and through the bore of the workpiece mounted on the table. The cutter bar shall be reciprocated by a drive mechanism conforming to 3.4.5 and shall cut on the "pull" stroke of the cutter.

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3.4.5 Cutter bar drive mechanism.

3.4.5.1 Type I (Mechanical) machines. Type I machines shall have a mechanical type cutter bar drive mechanism consisting essentially of an electric motor, reduction drive system, a rotating crank and a connecting rod that reciprocates a crosshead which slides on a vertically mounted slide mounted in the base of the machine. The crosshead shall have a clamp that securely attaches all cutter bars required to cut the full range of keyseats specified in Table I. Unless otherwise specified (see 6.2.1), the crosshead or slide shall have means for adjustment to compensate for wear in the mating sliding surfaces. The drive mechanism shall provide the cutting speeds, the number of cutting speed selections, and the keyseat lengths specified in Table I.

3.4.5.2 Type II (Hydraulic) machines. Type II machines shall conform to the requirements of 3.4.5.1 except that the drive system shall be hydraulically operated in lieu of mechanically operated. The hydraulic system shall consist of a hydraulic fluid reservoir, an electric motor driven pump, all necessary controls, piping, tubing, hoses, and a hydraulic piston and cylinder assembly which reciprocates the crosshead. In addition, hydraulically operated machines shall have an infinite number of cutting speed selections and shall have means of infinitely varying the stroke length. On Type II machines, the return speed of the cutter shall be not less than 80 percent greater than the fastest cutting speed.

3.4.6 Feed mechanism. All machines shall have a manual feed mechanism conforming to 3.4.6.1. Type I, Class 2, and Type II machines shall, in addition, have a power feed mechanism conforming to 3.4.6.2. The feed mechanism of Type I machines shall either feed the worktable toward the cutter or the cutter toward the worktable. The feed mechanism of Type II machines shall feed the cutter toward the workpiece. The feed mechanism shall have an adjustable stop device that will disengage or stop the feed mechanism after the keyseat has been cut to a preselected depth. The total feed distance shall conform to the requirements of Table I.

3.4.6.1 Manual feed system. Type I, Class 1 machines having a table feed system may have either a lever, pinion and rack type manual feed mechanism or a feed screw mechanism. All other machines shall have a manual feed system utilizing a feed screw having a dial graduated in increments not larger than 0.001 inch that indicates the feed distance.

3.4.6.2 Power feed system. Type I, Class 2 machines shall have a power feed system that is either of the incremental type or the "constant force" type. Type II machines shall have a power feed system of the incremental type. Incremental feed systems shall feed the cutter at a definite preselected rate during each return stroke of the cutter. On machines

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having an incremental power feed system, the mechanism shall provide the number of feed selections and feed range specified in Table I. Machines utilizing a constant force power feed system shall have a pneumatic piston and cylinder assembly that exerts a constant feed force on the cutter or table, as applicable. The feed force shall be infinitely variable over the range necessary to cut keyseats having widths conforming to the requirements of Table I. Constant force systems shall be designed to operate on a 85-100 psi input and shall be complete with air filters, pressure regulators, lubricators, control valves, piston and cylinder assembly, and necessary piping, tubing, and hoses. The system shall have a pressure gauge, calibrated either in "pounds per square inch" or "pounds, feed force" for monitoring and regulating the feed force.

3.4.6.3 Relief mechanism. All machines shall have a cutter relief mechanism that automatically moves the cutter away from the workpiece during the return stroke of the cutter to prevent marring of the workpiece and dulling of the cutter.

3.4.7 Cutter bar support. The machine shall have means of supporting the cutter bar to prevent side deflection and to prevent the cutter bar from springing away from the workpiece during the cutting stroke. The cutter bar support shall be of the center post type (see 3.4.7.1) or the resistance arm type (see 3.4.7.2).

3.4.7.1 Center post. On machines equipped with a center post type cutter bar support, the center post shall have a machined and ground slot running through its entire length. The cutter shall extend through the center post slot and shall reciprocate therein. The center post shall support the cutter bar and shall effectively prevent side deflection and tool springback. Center posts that will accommodate cutter bars that will cut the entire range of keyseat widths specified in Table I shall be available from the manufacturer. Necessary center post adaptors that will adapt the size of the center post to the size of the center post hole in the worktable shall be furnished with each center post ordered with the machine.

3.4.7.2 Resistance arm. On machines equipped with a resistance arm type cutter support, the arm shall be adjustable vertically and horizontally to accommodate various workpiece heights and various cutter bars. Means shall be provided to securely clamp the arm in any adjusted position. The resistance arm shall have a hardened roller or hardened replaceable insert that contacts the back of the cutter and prevents cutter springback. The table shall also have a slotted bushing, hardened roller or hardened replaceable insert at, or within 1 inch below, the table surface to prevent tool deflection at the table area.

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3.4.8 Taper keyseating mechanism. When specified (see 6.2.1), the machine shall be capable of cutting tapered keyseats. This may be accomplished through the use of a tilting table, an auxiliary tilting table that is attached to the main table, tapered center posts or taper wedges. The taper cutting range, in degrees or inches per foot, shall be as specified (see 6.2.1).

3.4.9 Indexing table. When specified (see 6.2.1), an index table shall be furnished to facilitate the machining of multiple keyseats or splines. When required, the indexing table shall provide the number of divisions specified (see 6.2.1).

3.4.10 Hydraulic system. The hydraulic system, when furnished, shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.11 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI C113.1. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI C113.1 for the disconnect fuse rating.

3.4.11.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements of Table I. Where two horsepower ratings are shown in Table I for Type II machines, the horsepower of the motor to be furnished shall be as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.5 Sizes and capacities. The sizes and capacities of the machine shall meet or exceed the requirements of Table I. The machine shall be capable of accommodating and securely clamping to the table any solid round workpiece having the sum of the workpiece radius plus bore radius, workpiece height and workpiece weight conforming to the requirements of Table I.

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TABLE I
SIZE AND CAPACITIES

Characteristic	Type I, Mechanical Machines				Type II, Hydraulic Machines			
	Class 1 Manual Feed		Class 2, Power Feed					
	Size 3/4	Size 1½	Size 1½	Size 1½	Size 1½	Size 2	Size 3	Szie 3½
Size keyseat that can be machined:								
Width range, inches	1/16-3/4	1/16-1½	1/16-1½	1/16-1½	1/16-1½	1/4-2	1/4-3	1/4-3½
Length, maximum, inches	7	17	9	18	12	24	24	30
Cutting speeds:								
Number of cutting speed selections	1	1	1	1	infinitely variable			
Cutting speed, Type I machines strokes/min	48	42	34	42	-	-	-	-
Cutting speed range, Type II machines:								
Lowest speed, ft/min	-	-	-	-	0 to 9	0 to 13	0 to 12	0 to 10
Spread between lowest to highest speed, ft/min	-	-	-	-	12	8	18	20
Feed data:								
Number of power feed selections	none	none	3	inf. var feed force not incremental	inf. var.	10 or inf. var.	10 or inf var.	inf. var.
Power feed range, inch/stroke	none	none	0.002-0.006		0.002-0.010	0.002-0.010	0.002-0.010	0.002-0.010
Total feed distance, inches	1-3/8	1-3/8	1-3/8	1½	1-3/8	1-3/8	1-9/16	1-13/16
Workpiece size:								
Sum of workpiece radius plus bore radius, inches	15	21	16	22	not applicable	not applicable	34	34
Workpiece height, inches	8	18	14	18	14	25	25	32
Workpiece weight, pounds	1,500	2,000	1,500	2,000	5,000	10,000	20,000	20,000
Horsepower, cutter drive motor	1½	2	2	2	3 or 5 (see 3.4.11.1)	5 or 7½ (see 3.4.11.1)	10	10 or 15 (see 3.4.11.1)
Depth of pit required to install machine, inches, not more than	0	0	0	0	0	32	42	56

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

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3.6 Performance and accuracy.

3.6.1 Keyseat sizes versus bore sizes. The machine shall be capable of machining all keyseats (up to the maximum widths specified in Table I) in bore diameters within the range specified in Table II.

TABLE II

KEYSEAT SIZE VERSUS BORE SIZE

Keyseat Size (Inches)			Nominal Bore Diameter Range (Inches)	
Width	Nominal Depth		Over	Up To and Including
	Square	Rectangular		
3/32	3/64	-	5/16	7/16
1/8	1/16	3/64	7/16	9/16
3/16	3/32	1/16	9/16	7/8
1/4	1/8	3/32	7/8	1-1/4
5/16	5/32	1/4	1-1/4	1-3/8
3/8	3/16	1/8	1-3/8	1-3/4
1/2	1/4	3/16	1-3/4	2-1/4
5/8	5/16	7/32	2-1/4	2-3/4
3/4	3/8	1/4	2-3/4	3-1/4
7/8	7/16	5/16	3-1/4	3-3/4
1	1/2	3/8	3-3/4	4-1/2
1-1/4	5/8	7/16	4-1/2	5-1/2
1-1/2	3/4	1/2	5-1/2	6-1/2
1-3/4	7/8	3/4	6-1/2	7-1/2
2	1	3/4	7-1/2	9
2-1/2	1-1/4	7/8	9	11
3	1-1/2	1	11	13
3-1/2	1-3/4	1-1/4	13	15

Example: Machine shall be capable of machining a 1/2 by 1/4 square keyseat or a 1/2 by 3/16 rectangular keyseat in all bores ranging in size from over 1-3/4 inch diameter to, and including, 2-1/4 inch diameter.

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3.6.2 Accuracy requirements. The machine shall be capable of cutting keyseats in steels having physical properties equal to or exceeding those of AISI 4140 carburizing type chrome-moly steel. The finished keyseats shall have width and depth tolerances conforming to Class 2 fit requirements as defined in Table 5 of ANSI B17.1. In addition, the offset error, lead error, and parallelism shall conform to the keyseat alignment tolerances in Section 7 of ANSI B17.1. These requirements are without regard to metal removal rate.

3.6.3 Metal removal rate requirements. Size 3/4, 1-1/4, and 1-1/2 machines shall be capable of cutting a 3/4-inch wide by 3/8-inch deep keyseat in AISI 4140 steel in not more than 1-1/4 minutes per inch of keyseat length. Size 2, 3 and 3-1/2 inch machines shall be capable of cutting a 2-inch wide by 1-inch deep keyseat in AISI 4140 steel in not more than 6 minutes per inch of keyseat length. While machining at these metal removal rates, the finished keyseat shall conform to, or exceed, Class 1 fit requirements as defined in Table 4 of ANSI B17.1, and the offset error, lead error, and parallelism shall conform to Section 7 of ANSI B17.1.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. Due to the variety of machine designs covered by this specification, it is impractical to list, in detail, the specific equipment that shall be furnished by the supplier. The procuring agency will specify in 6.2.1.p. the range of keyseat widths and lengths, the workpiece diameters and heights, and the bore diameters that the user is planning to machine. The supplier shall furnish with each machine all necessary equipment such as centering devices, hold-down devices, cutter bars, tooth inserts, center posts, bushings and adaptors applicable to the design of his machine and which are necessary to meet the size range requirements designated by the procuring agency. To facilitate and expedite acceptance of the machine by the using agency, the machine supplier shall certify that all necessary equipment is furnished, and shall identify all equipment as to its usage and size range.

3.8 Optional equipment. Optional equipment and tooling shall be furnished as specified and fully described (see 6.2.1).

3.9 Additional requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified (see 6.2.1).

3.10 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

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3.11 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.12 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input (volts, total amps, phase, frequency)

Maximum keyseat width and length

Contract Number or Order Number

National Stock Number or Plant Equipment Code

Date of manufacture

U S

3.13 Technical data. Date shall be furnished as specified (see 6.2.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.2.3, and the inspection in 4.6. Failure of an item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and its equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.13.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.5.2 Performance tests. Each machine shall be subjected to the following performance tests:

4.5.2.1 Workpiece. The workpiece for the performance tests shall consist of a gear blank having a machined bore and a thickness large enough for cutting the length and width keyseat specified in 4.5.2.2. The face of the workpiece that rests on the worktable shall be finish machined to provide a true mounting surface. The workpiece shall have physical properties equal to, or exceeding, those of AISI 4140 chrome-moly steel. Scrap material may be utilized if it conforms to the above requirements.

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4.5.2.2 Accuracy test. Using the workpiece specified in 4.5.2.1, the machine shall cut a keyseat having a width not less than 75 percent of the maximum rated keyseat width and a length not less than 25 percent of the maximum rated keyseat length as specified in Table I. The cutting speed and feed rate shall be at the option of the manufacturer. The width and depth tolerance, offset error, and parallelism of the finished keyseat shall conform to the requirements of 3.6.2.

4.5.2.3 Metal removal rate test. The workpiece used in the test specified in 4.5.2.2 shall be indexed to a new position. Size 3/4, 1-1/4, and 1-1/2 machines shall be used to cut a nominal 3/4-inch wide by 3/8-inch deep keyseat. Size 2, 3, and 3-1/2 inch machines shall cut a nominal 2-inch wide by 1-inch deep keyseat. The time required to machine the keyseat, the width and depth tolerances, the offset error, lead error, and parallelism shall conform to the requirements of 3.6.3.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. Machines covered by this specification are intended to be used for cutting keyseats for square and flat parallel side keys and tapered keys in the bores of gears, propellers, or similar parts.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number and date of this specification.

b. Type, class and size required (see 1.2).

c. When a first article is required for inspection and approval (see 3.1, 4.2 and 6.3).

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d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Specify additional safety and health requirements (see 3.2.4).

h. Painting, if different (see 3.3.5).

i. Centering device capable of centering workpieces having outside diameters that are non-concentric with bores, if required (see 3.4.3).

j. Crosshead or slide adjustment to compensate for wear, if not required (see 3.4.5.1).

k. Taper cutting mechanism, if required; specify required taper (see 3.4.8).

l. Indexing table, if required; specify number of divisions required (see 3.4.9).

m. Electrical equipment, if different (see 3.4.11).

n. Specify desired horsepower for Type II machines selected from Table I (see 3.4.11.1).

o. Type of motor enclosure, if different (see 3.4.11.1).

p. Standard equipment, if different; ranges of keyseat widths, lengths, workpiece diameters, workpiece heights and bore diameters which the user plans to machine. (Supplier will furnish all necessary equipment such as centering devices, hold-down devices, cutter bars, tooth inserts, center post, bushings, and adaptors to machine the ranges specified here (see 3.7).

q. Optional equipment, if required; full describe required optional equipment (see 3.8).

r. Specify additional features required (see 3.9).

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s. Type fungus proofing, if required (see 3.10).

t. Lubrication chart or plate, if different (see 3.11).

u. Nameplate, if different (see 3.12).

v. First article inspection, if different (see 4.2).

w. Quality conformance inspection, if different (see 4.3).

x. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requiring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the requiring agency should specify the exact configuration of the guard required as in most cases the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Air Force - 99

Review Activities:
Army - AL, ER

User Activities:
Army - SM, AR
Navy - MC

Preparing Activity:
DLA - IP

Project Number:
3419-0199

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-K-80221A	Keyseating Machine, Mechanical and Hydraulic Reciprocating Cutter Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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